A singleton omission orbit at an interior crossing

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$ be a faithful transitive string group generated by involutions, $n \geq 8$, with every maximal omission intransitive. Suppose the unique eligible i-edge $\{a, b\}$ has $0 < i < r-1$, and the two G_i-orbits are A and the singleton $\{b\}$. Write $L = G_{<i}$, $R = G_{>i}$ and $d = r - 1 - i$. All generators except ρ_i fix b; those at distance more than one from i also fix a, by the unique-crossing lemma. No IP for a quotient action will be used.

Proposition 1. *Under these hypotheses, $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Neither L nor R is transitive on A. Indeed, if L were, then $\langle L, \rho_i \rangle$ would be transitive on $A \cup \{b\}$, contradicting intransitivity of any omission with index greater than i. The argument for R is the dual one. Both factors are nontrivial, because the interior index leaves generators on both sides and a redundant generator would have a transitive omission.

Since L and R commute and $G_i = LR$ is transitive on A, the L-orbits form $m \geq 2$ blocks of common size $k \geq 2$. The action of L on any one of these blocks is faithful: its kernel fixes all R-translates of that block, hence all of A and also b. Thus its i distinguished generators retain independence in degree k. Whiston gives $k \geq i + 1$.

Every distinguished R block image is essential. Otherwise the remaining R images, together with L, would still be transitive on A. The omitted generator is supported on A and so cannot join any orbits of that global omission. Adjoining it produces transitive G, so the omission would already be transitive, a contradiction.

In particular the last block image is nontrivial. The distance to a moved-block argument gives $m \geq d+1$: start from $\beta = a^L$, and use a shortest path of earlier R block images to a block moved by the last generator. Every intervening index must occur, by commuting the factors on opposite sides of a missing index; the upper factors fix β and the lower factors commute with the last generator. The path has at least $d-1$ edges, and its last edge under ρ_{r-1} gives a further block. For $d = 1$ the same count is just $m \geq 2$. Consequently

$$n = km + 1 \geq (i+1)(d+1) + 1 = 2r + (i-1)(d-1) - 1. \tag{1}$$

If $i, d \geq 2$ and $(i-1)(d-1) \geq 2$, (1) gives $n \geq 2r + 1$. The remaining case with $i, d \geq 2$ is $i = d = 2$, so $r = 5$. Rank failure would force $n \leq 10$, while (1) gives $n \geq 10$; hence $k = m = 3$ and $n = 10$. But each of the two distinguished involutions of L is a transposition on each faithful three-point orbit. It is then a product of three transpositions and fixes b, contrary to evenness.

We may therefore assume one of i, d is one. Reverse the string if necessary, so $i = 1$. Then $L = \langle \rho_0 \rangle$ has two-point orbits throughout A: $k = 2$, $n = 2m + 1$, and ρ_0 exchanges all m pairs. Its evenness forces m even. In particular $n \geq 8$ gives $m \geq 4$. If the rank bound fails, $r = d + 2 > m$, while $m \geq d + 1$. Thus $d = m - 1$ and $r = m + 1$.

The essential R block images form an independent family of $m - 1$ involutions in S_m, so their image is S_m, by Whiston's equality clause. As already proved, R is not transitive on A. Its transitive action on the m pairs then forces exactly two point orbits U, V, each meeting each pair once. The block map on R is faithful, since a kernel element fixes the one point in each pair belonging to each of U, V. Thus R is the diagonal natural S_m on these two rows.

Let $a' = a^{\rho_0}$ and put $H = \langle \rho_3, \ldots, \rho_{r-1} \rangle$. It fixes a by unique crossing, and a' by commutation with ρ_0. Its $m - 2$ block images remain independent, fix the pair $\{a, a'\}$, and hence generate

the full S_{m-1} on the remaining pairs, again by Whiston. In particular H has only the two fixed points a, a' in A, and is diagonal natural S_{m-1} on the remaining two rows.

The crossing involution is $s = \rho_1 = (a\ b)u$, where u is supported on $A\setminus\{a\}$. Since H fixes a, b and commutes with s, the permutation u centralizes H. It fixes a, so it fixes a' as well. On the remaining two rows, each of size $m-1 \geq 3$, the centralizer of the diagonal natural S_{m-1} consists of the identity and the unique equivariant row exchange. Indeed a commuting permutation permutes its two orbits, and the natural symmetric action has trivial centralizer. The exchange here is the restriction of ρ_0, so is the product v of the other $m-1$ pair transpositions. Thus $u \in \{1, v\}$. Since s is even, $u = v$, which is odd because m is even.

But s now joins the two R-orbits U, V through any one of these other pairs, and joins b to a. Consequently $G_0 = \langle R, s\rangle$ is transitive, the final contradiction. This proves the bound in every case. □

Scope and attribution. This proves the interior singleton-orbit branch used from Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.25, physical page 33. The source's commuting-block reduction and the retained Whiston bound are reused; the actual-orbit count and diagonal-row omission argument replace its path-disjointness and equality-graph steps. The unique-crossing input is paid in full. The moved-block argument is reproduced here so no nontrivial-second-orbit hypothesis is imported from the other support note. No maximal-string classification, finite census, or projected IP is needed. The endpoint indices $i = 0, r-1$ are explicitly outside this statement. There is no full L044, new question, formalization or priority claim.